Amir Nourinia

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ABOUT ME

Systems & Graphics Software Engineer | Medical Imaging & AI Specialist
Innovative software engineer with a deep expertise in systems and graphics programming. Passionate about advancing medical imaging through AI-driven solutions, I excel at designing and implementing high-performance, scalable software architectures that bridge the gap between cutting-edge technology and healthcare innovation.

EDUCATION AND TRAINING

09/2006 – 06/2009 Tabriz, Iran
HIGH SCHOOL DIPLOMA Madani High School - National Organization of Exceptional Talents (NODET)

09/2009 – 06/2010 Tabriz, Iran
PRE-UNIVERSITY DIPLOMA Madani High School - National Oranization of Exceptional Talents (NODET)

09/2010 – 11/2015 Tehran, Iran
ELECTRICAL ENGINEERING - BIOELECTRICS (B.SC.) Amirkabir University of Technology

Field of study Electrical Engineering | **Final grade** 15.98/20 | **Thesis** Smart MRA injection Robot

09/2016 – 04/2023 Munich, Germany
COMPUTER SCIENCE - BIOMEDICAL COMPUTING (M.SC.) Technical University of Munich

Field of study Computer Science and Biomedical Computing | **Final grade** Completed all coursework; thesis incomplete.

PUBLICATIONS

2016
Design and fabrication of a nanofibrous polycaprolactone tubular nerve guide for peripheral nerve tissue engineering using a two-pole electrospinning system

2020

Orientation of directional deep brain stimulation leads on CT: Resolving the ambiguity

WORK EXPERIENCE

07/2023 – 10/2024 Munich, Germany
SENIOR SOFTWARE ENGINEER HOLO-LIGHT GMBH

Systems & Graphics Software Engineer
Key Responsibilities:
- System Design & Architecture: Led XR system design with an emphasis on low-latency streaming.
- OpenXR Library Development: Designed and architected a cross-platform OpenXR library.
- Android Development: Utilized NDK, Java/Kotlin, and MediaCodec for Android XR apps.
- Unity Development: Created immersive 3D GUIs using Unity, focusing on AR/VR.
- WebRTC Streaming: Developed a WebRTC-based service for real-time communication.
- Implemented gRPC-based communication protocols between microservices to enable efficient, low-latency data exchange in a distributed architecture
- Graphics Programming: Implemented OpenGL ES, Vulkan, and GLSL for optimized XR graphics.
- Team Support & Mentorship: Guided the team on C++, algorithms, and clean code principles.

Skills:
- Programming Languages: C++, C#, Java, Kotlin
- Graphics & Shading: OpenGL, OpenGL ES, Vulkan, GLSL
- XR & VR Development: OpenXR, Unity, MRTK3
- Networking & Streaming: WebRTC, TCP/UDP, P2P Streaming

- System Development: NDK, JNI, Android Development, MediaCodec
- Software Architecture: Designed cross-platform libraries; taught clean code practices

08/2022 – 03/2023 Munich, Germany
WORKING STUDENT - SOFTWARE DEVELOPER ONE-PROJECTS GMBH

C++ Systems Programming and OOP Python library development - Working Student
Key Responsibilities:
- C++ systems programming
- Qt Developement (C++/Qt)
- imgui Developement (C++/OpenGL)
- Improvements over Conan build system based on python
- server load control scripts based on python

02/2022 – 30/04/2022 Munich, Germany
WORKING STUDENT - SOFTWARE DEVELOPER AST (ADAPTIVE SENSORY TECHNOLOGY) GMBH

Systems Programming (Graphics Programming i.e. OpenGL, Cross Compilation Toolchain) for ARM64 based Embedded Devices - Working Student
Key Responsibilities:
- C++ systems programming
- Graphic Shader Programming (GLSL/HLSL)
- Cross Compilation pipeline setup using LLVM and CMake
- Remote access network infrastructure setup

04/2021 – 30/06/2021 Munich, Germany
WORKING STUDENT - SOFTWARE DEVELOPER FIREFLOW GMBH

iOS Development, Swift Programming
Key Responsibilities:
- iOS development using SwiftUI
- Refactoring and Documenting Legacy iOS code
- Improving build systems
- Interacting with users and fixing bug reports

08/2020 – 02/2021 Munich, Germany
WORKING STUDENT - SOFTWARE DEVELOPER SURGEVISION GMBH

Software Engineering and Algorithm Development for Medical Image Processing - Working Student
Key Responsibilities:
- Qt Development (C++/Qt)
- Test Driven Computer Vision Algorithm Development (C++/OpenCV, GoogleTest)
- Setting up Jenkins based CI/CD pipeline

08/2019 – 03/2020 Munich, Germany
WORKING STUDENT - SOFTWARE DEVELOPER BRAINLAB GMBH

Algorithm Development for Medical Image Processing Using C++ and Python - Working Student
Key Responsibilities:
- Research
- Computer Vision Algorithm Development (C++)
- Computational Modeling (C++)
- Data Visualization (Python, C++)

INTERESTS

Numerical Programming

Algorithms

Neuroscience

Computer Vision

Machine Learning

Computer Graphics Programming

Software Engineering

PROGRAMMING LANGUAGES AND TECHNOLOGIES

Systems Programming

My main systems programming language is Modern C++ (C++23). Although, I am familiar with Rust and Swift and I have used them for several projects in web assembly and iOS.

Python World

I have used OOP type annotated Python extensively for University projects and for my previous working experiences. In my past experiences, I have used different python libraries in the scientific computation realm (Scipy, PyTorch, etc.) to solve computer vision and machine learning problems. Also, I have used python for backend development using frameworks such as Django and Flask.

Front-End

JavaScript/TypeScript: React, Electron
C++: Qt, ImGui
Swift: SwiftUI, UIKit
Java/Kotlin: JavaFX, Android Native components, Jetpack Compose

Unreal Engine

I have used Unreal Engine for developing handheld Augmented Reality Applications for university projects.

Unity Engine

I have used Unity Engine extensively during my work at Hololight GmbH for developing Front-end Augmented Reality and Mixed Reality Applications for Head Mounted platforms such as Meta Quest, Magic Leap 2, Apple Vision Pro and Microsoft HoloLens 2.

SELECTED COURSES

Linear Control Systems

Micro-controller Programming

Computer Vision

Computational Intelligence

C++ Programming

Selected Topics in Magnetic Resonance Imaging

Signals and Systems Analysis

Robotics

Physiology

Systematic Human Anatomy

LANGUAGE SKILLS

Mother tongue(s): **AZERBAIJANI** | **PERSIAN**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
TURKISH	C2	C2	C1	C2	C2
ENGLISH	C2	C2	C2	C2	C2
GERMAN	B1	B1	B1	B1	B1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

HOBBIES AND INTERESTS

Basketball

Hiking